Guest speaker

Andreas Pasternak

Scaling Is All You Need: Autonomous Driving with JAX-Accelerated Reinforcement Learning

Cornell RLRS x Doordash AI Labs

DOORDASH

Speaker

Our guest

Principal Engineer at Doordash Labs
DoorDash · Full-time
Nov 2024 - Present · 1 yr 10 mos

RL Algo and Data Lead
Zoox · Full-time
May 2023 - Nov 2024 · 1 yr 7 mos

Senior Staff Software Engineer
Cruise · Full-time
Feb 2022 - May 2023 · 1 yr 4 mos

Andreas Pasternak

Principle Engineer @ DoorDash Labs

Before joining DoorDash, Andreas spent four years at Zoox, where he led the RL algorithms team working on autonomous driving (with 2 patents!) as well as Cruise. His past work includes:

TITLE
Adversarial agent controls generation and problematic scenario forecasting M Kobilarov, JB Packer, G Garimella, A Pasternak, Y Zhang, R Yu US Patent 11,891,088
Scaling is all you need: Autonomous driving with jax-accelerated reinforcement learning M Harmel, A Paras, A Pasternak, N Roy, G Linscott arXiv preprint arXiv:2312.15122
Automated reinforcement learning scenario variation and impact penalties G Linscott, A Pasternak, JB Packer, M Kobilarov US Patent 12,037,013

and currently is building Dot (Doordash Autonomous Delivery Team) with model based RL

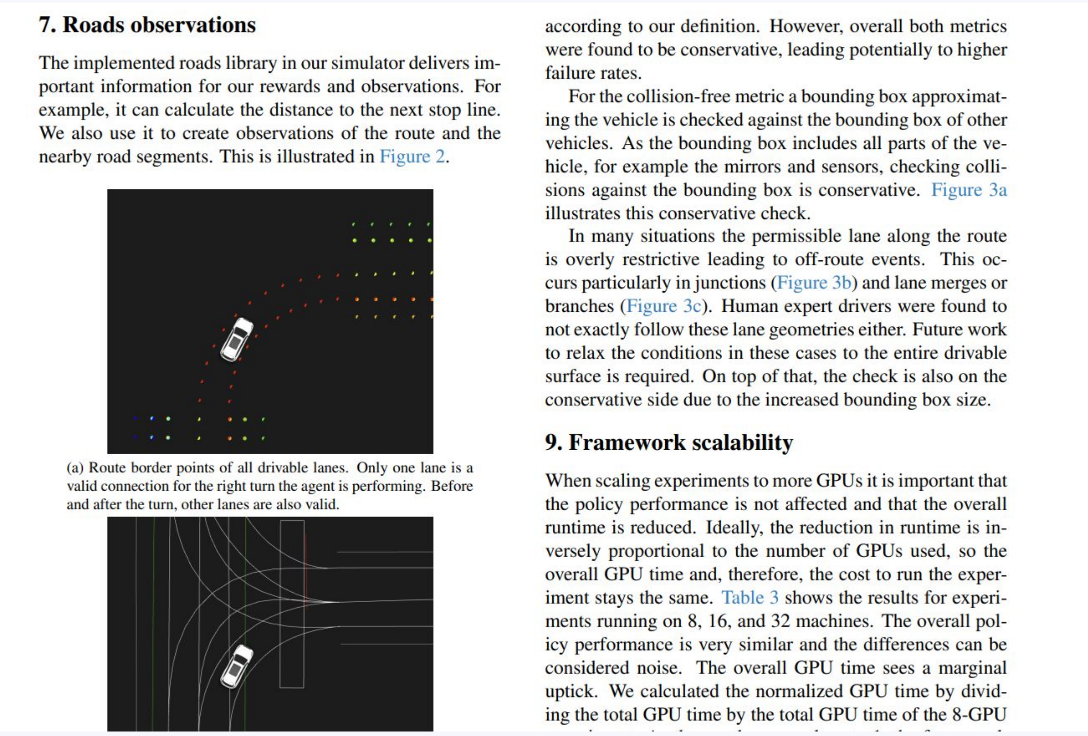
RL algorithms | Autonomous driving | JAX | 2 patents

Paper

This week's reading

Paper · arXiv 2312.15122

Scaling Is All You Need: Autonomous Driving with JAX-Accelerated Reinforcement Learning



7. Roads observations

The implemented roads library in our simulator delivers important information for our rewards and observations. For example, it can calculate the distance to the next stop line. We also use it to create observations of the route and the nearby road segments. This is illustrated in Figure 2.

(a) Route border points of all drivable lanes. Only one lane is a valid connection for the right turn the agent is performing. Before and after the turn, other lanes are also valid.

according to our definition. However, overall both metrics were found to be conservative, leading potentially to higher failure rates.

For the collision-free metric a bounding box approximating the vehicle is checked against the bounding box of other vehicles. As the bounding box includes all parts of the vehicle, for example the mirrors and sensors, checking collisions against the bounding box is conservative. Figure 3a illustrates this conservative check.

In many situations the permissible lane along the route is overly restrictive leading to off-route events. This occurs particularly in junctions (Figure 3b) and lane merges or branches (Figure 3c). Human expert drivers were found to not exactly follow these lane geometries either. Future work to relax the conditions in these cases to the entire drivable surface is required. On top of that, the check is also on the conservative side due to the increased bounding box size.

9. Framework scalability

When scaling experiments to more GPUs it is important that the policy performance is not affected and that the overall runtime is reduced. Ideally, the reduction in runtime is inversely proportional to the number of GPUs used, so the overall GPU time and, therefore, the cost to run the experiment stays the same. Table 3 shows the results for experiments running on 8, 16, and 32 machines. The overall policy performance is very similar and the differences can be considered noise. The overall GPU time sees a marginal uptick. We calculated the normalized GPU time by dividing the total GPU time by the total GPU time of the 8-GPU

Moritz Harmel · Anubhav Paras · Andreas Pasternak · Nicholas Roy · Gary Linscott

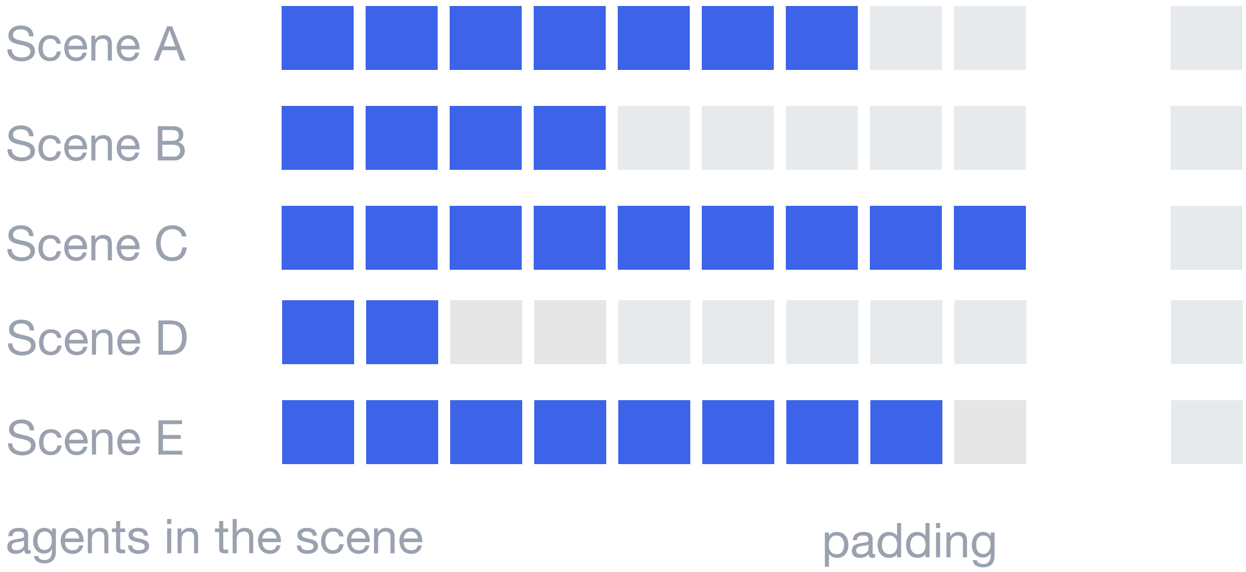
Zoox, Inc. · 2023

Data: Scenes from real driving

6000h

of human-expert driving in San Francisco, logged and replayed as simulation scenes

Every scene is a different size. All of them get padded to one.



Paper

Why autonomous driving is hard to scale

1 You cannot collect it on the road

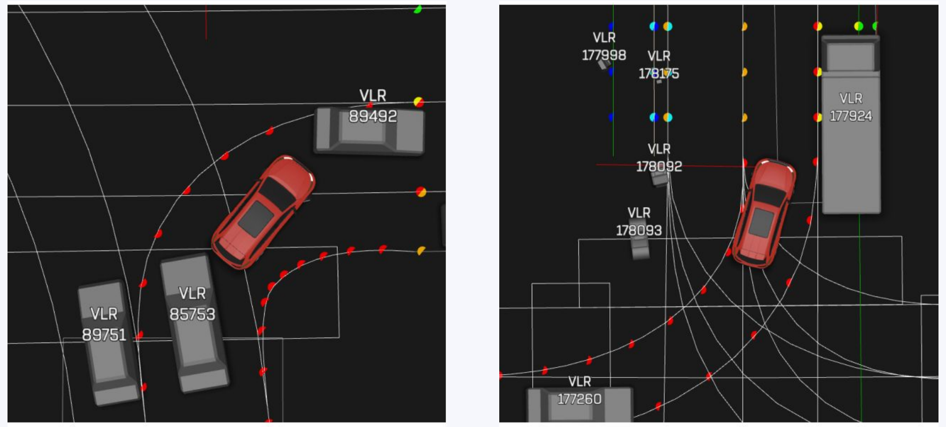
Gathering experience on real vehicles is neither safe nor scalable at the volume RL needs

2 The simulator is the bottleneck

Billions of environment steps only happen if a step costs well under a millisecond

3 One accelerator is not enough

Actors and learners have to run asynchronously across many GPUs to keep them fed



The claim

Remove those three limits and closed-loop driving performance improves monotonically with both model size and dataset size

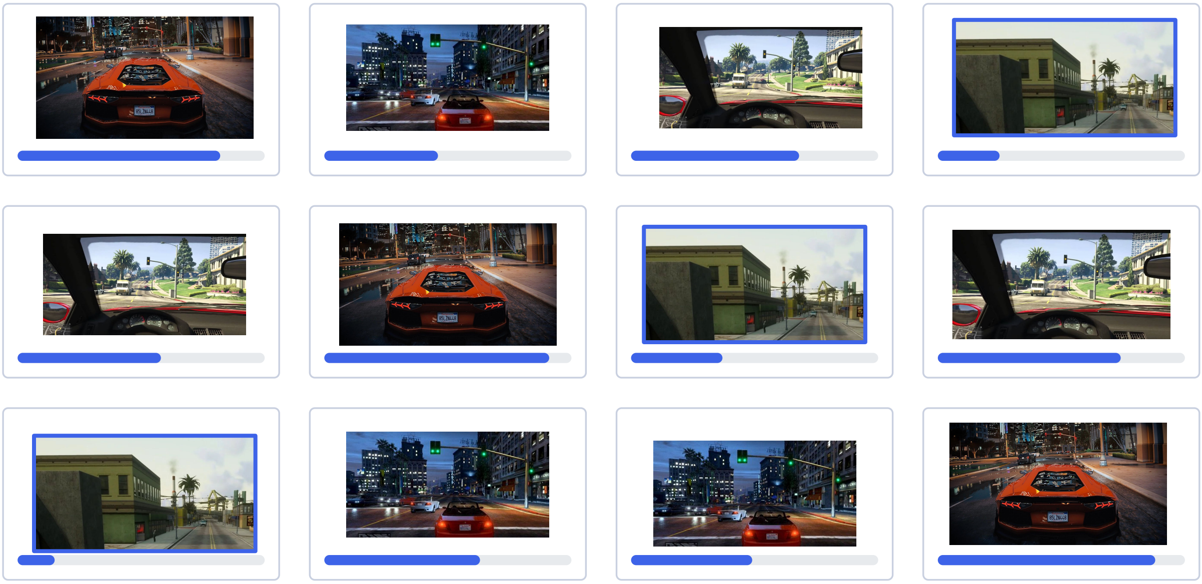
Paper

The training loop

updated weights back to the actors

Data	Environment	Actors	Learner
Logged driving	**JAX simulator**	**Collect rollouts**	**PPO + V-trace**
6000 h, San Francisco	jit + jax.lax.scan, on GPU	many, asynchronous (can be off-policy)	off-policy correction via reweighting

Actors



one per device, each mid-episode

Learner

$$v_s = V(x_s) + \sum_{t=s}^{s+n-1} \gamma_{t-s} \left(c_s \cdots c_{t-1} \right) \delta_t V$$

$$\delta_t V = \rho_t \left(r_t + \gamma V(x_{t+1}) - V(x_t) \right)$$

$$c_i = \min(\bar{c}, \pi/\mu)$$

$$\rho_t = \min(\bar{\rho}, \pi/\mu)$$

PPO + V-trace

consumes whatever has arrived

No stalling: actors push trajectories whenever they finish; the learner pulls whenever it is ready

Stays on device: rollout, reward and update all execute on the accelerator

Paper

The training loop

updated weights back to the actors

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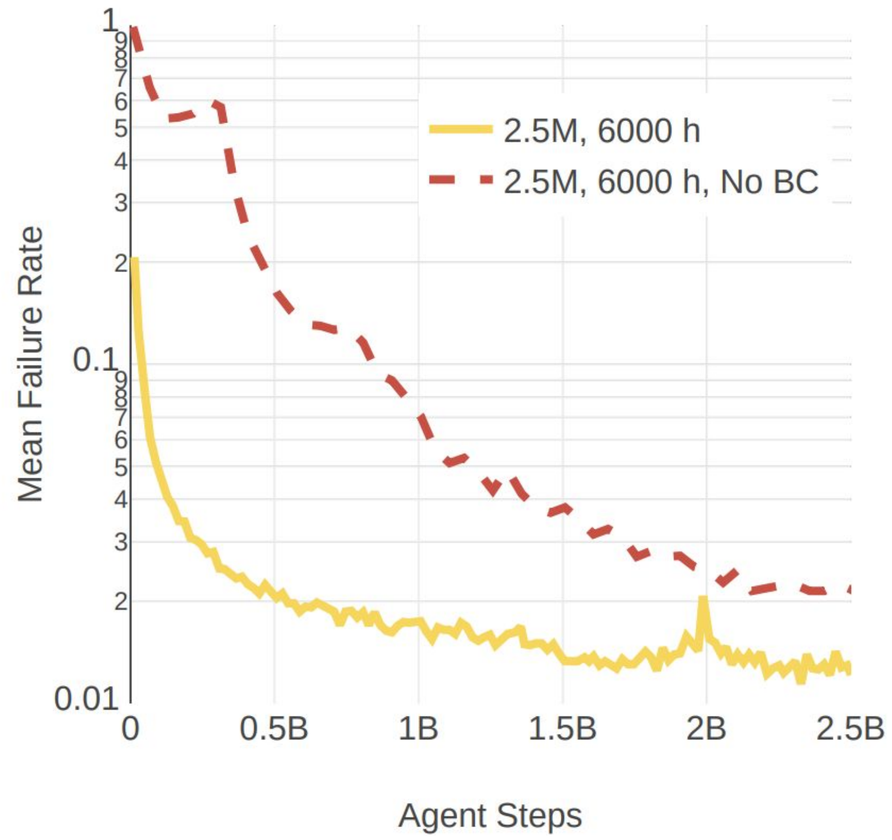
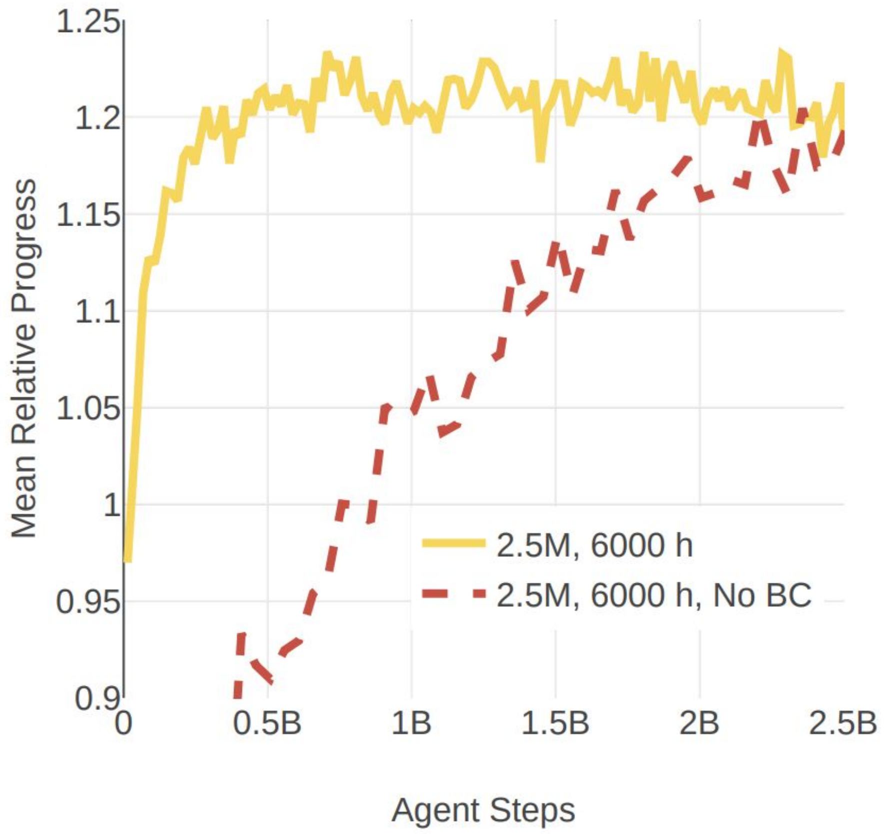
Training curves for experiments with and without the SL pre-training on the 2.5M parameter model on 2 metrics (failure rate and progress ratio):





Warm start: behaviour cloning on the same 6000 h, before PPO ever runs

Stays on device: rollout, reward and update all execute on the accelerator

Results

The scaling grid + SOTA

	600 h	2000 h	6000 h
0.75M	3.12%	2.27%	1.38%
2.5M	1.92%	1.19%	1.14%
25M	not run	1.35%	0.88%

Behaviour cloning, this paper 19.85%

MGAIL, best prior result 2.45%

BC + PPO, this paper (64% lower than the previous best) 0.88%

Rows: model parameters

Minimum failure rate falls along both axes, with no plateau inside the range they tested.

The corner cell (largest model + most data) has the best result in the paper

0.88%

failure rate, 25M parameters on 6000 h

Career Talk

Leading Research Teams in Industry